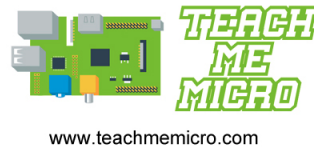
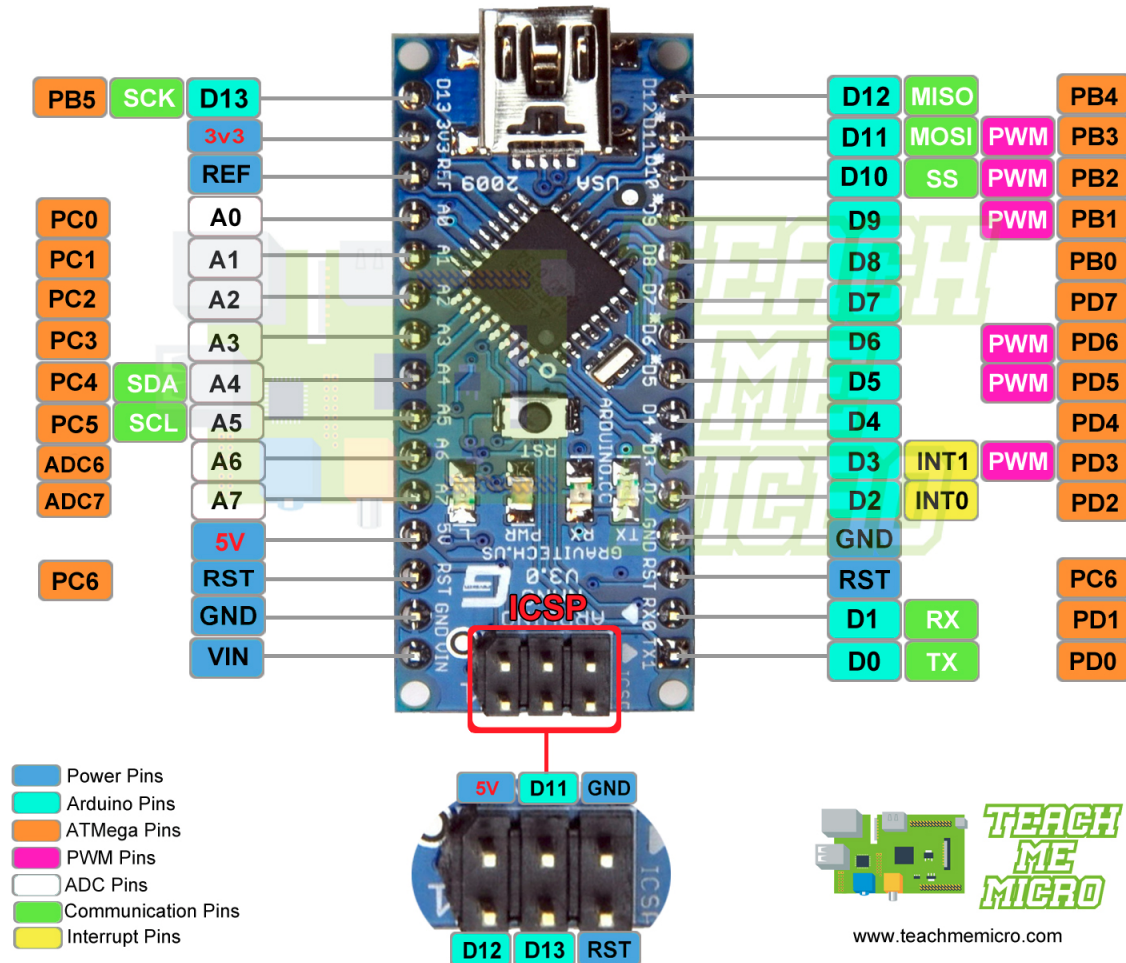


Line Following Robot

Voidash

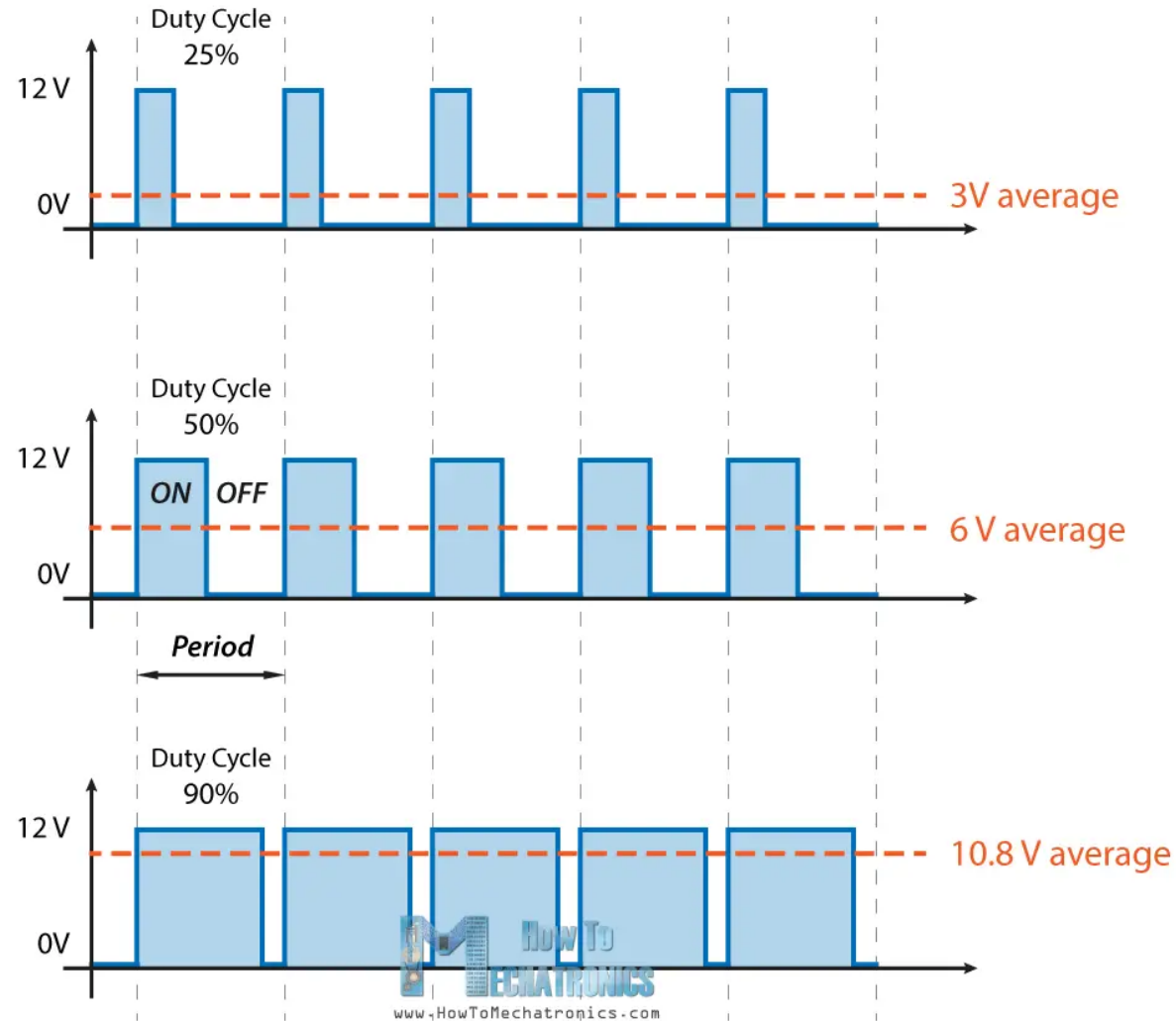
ARDUINO NANO PINOUT

Mini USB Port

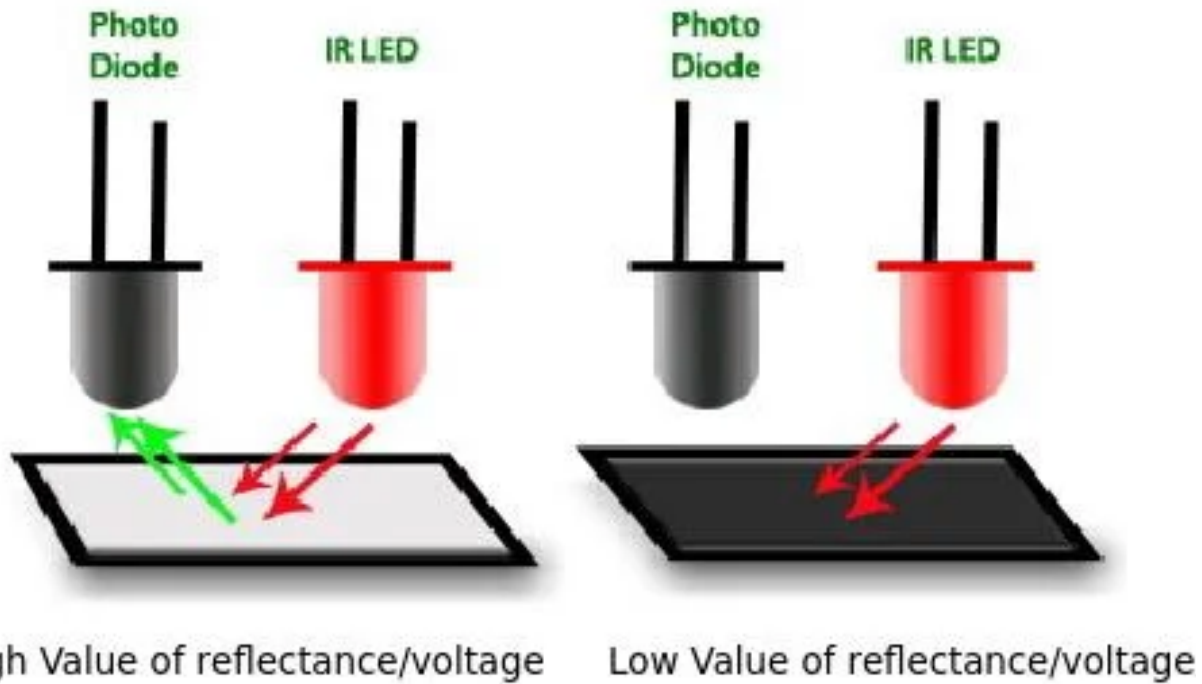


PWM (Pulse width modulation)

Pulse Width Modulation



Basics of IR Sensor



```

int lm = 5;
int l1 = 18;
int c = 19;
int r1 = 22;
int rm = 21;

int irPins[] = {lm,l1,c,r1,rm};

int pinRead[] = {0,0,0,0,0};
void setup() {

    pinMode(lm, INPUT);
    pinMode(l1, INPUT);
    pinMode(c, INPUT);
    pinMode(r1, INPUT);
    pinMode(rm, INPUT);

    // for debugging. The output will appear on the serial monitor
    // To open the serial monitor, click the magnafing glass icon in the upper right corner
    Serial.begin(115200);

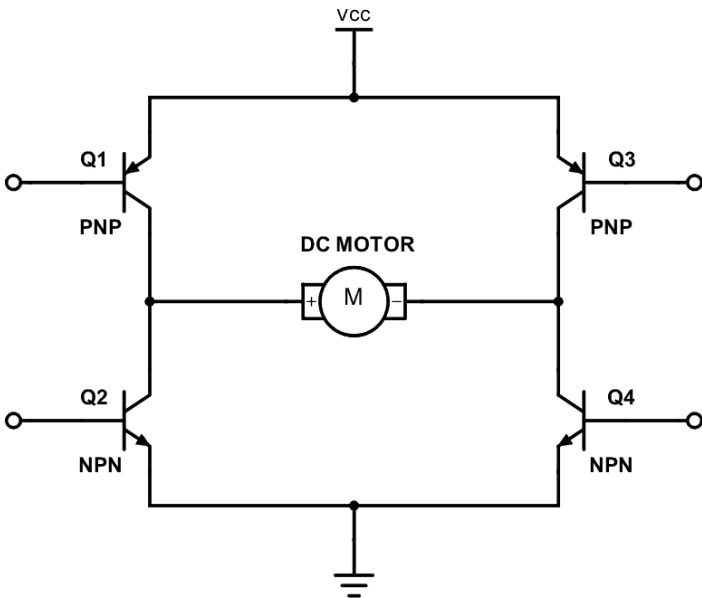
}

void loop() {

    for(int i = 0 ;i < 5; i++){
        pinRead[i] = digitalRead(irPins[i]);
    }
    for(int i = 0 ;i < 5; i++){
        Serial.print(pinRead[i]);
    }
    Serial.println();
    delay(300);

}

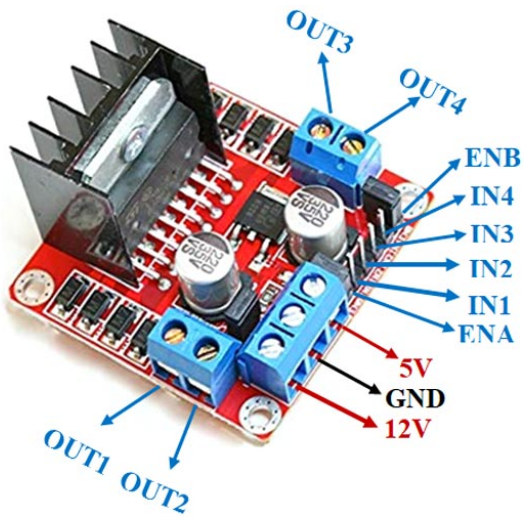
```



H Bridge

- controls the direction of rotation of DC motor, clockwise or anticlockwise
- if Q1 and Q4 are turned on then the motor moves clockwise
- if Q2 and Q3 are turned on then the motor moves anticlockwise

Motor Control



Pin Name	Description
IN1 & IN2	Motor A input pins. Used to control the spinning direction of Motor A
IN3 & IN4	Motor B input pins. Used to control the spinning direction of Motor B
ENA	Enables PWM signal for Motor A
ENB	Enables PWM signal for Motor B
OUT1 & OUT2	Output pins of Motor A
OUT3 & OUT4	Output pins of Motor B
12V	12V input from DC power Source

Motor test

```
int right_forward = 26;
int right_reverse = 27;
int left_forward = 33;
int left_reverse = 25;
int delay_time_on = 2000; // how long should each wheel turn?
int delay_time_off = 2000; // delay between tests

void setup() {
  // Turn these pins on for PWM OUTPUT
  pinMode(right_forward, OUTPUT);
  pinMode(right_reverse, OUTPUT);
  pinMode(left_forward, OUTPUT);
  pinMode(left_reverse, OUTPUT);
  // turn all the motors off
  digitalWrite(right_forward, LOW);
  digitalWrite(right_reverse, LOW);
  digitalWrite(left_forward, LOW);
  digitalWrite(left_reverse, LOW);
  // for debugging. The output will appear on the serial monitor
  // To open the serial monitor, click the magnafing glass icon in the upper right corner
  Serial.begin(115200);
}

void loop() {
  Serial.println("Right Forward Test");
  digitalWrite(right_forward, HIGH);
  delay(delay_time_on);
  digitalWrite(right_forward, LOW);
  delay(delay_time_off);

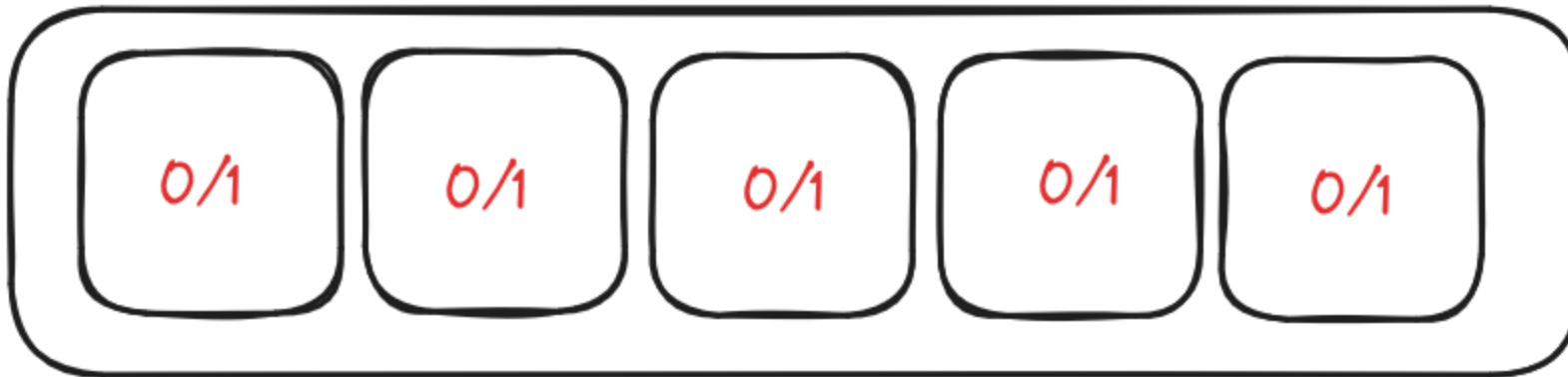
  Serial.println("Right reverse test");
  digitalWrite(right_reverse, HIGH);
  delay(delay_time_on);
  digitalWrite(right_reverse, LOW);
  delay(delay_time_off);

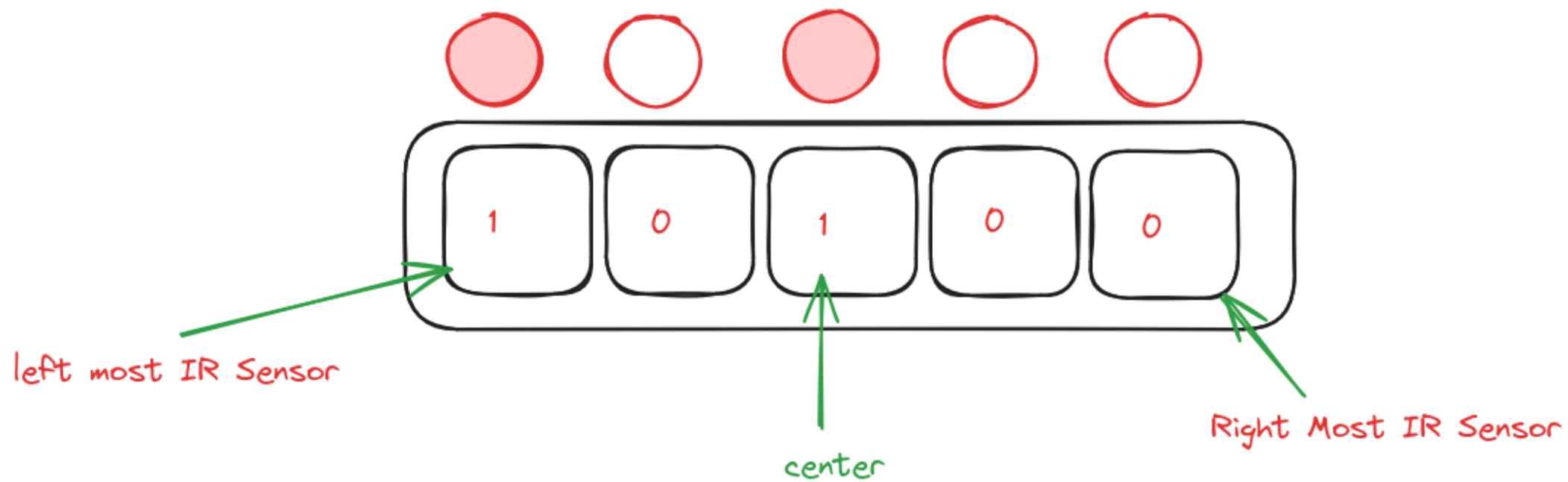
  Serial.println("Left Forward Test");
  digitalWrite(left_forward, HIGH);
  delay(delay_time_on);
  digitalWrite(left_forward, LOW);
  delay(delay_time_off);

  Serial.println("Left Reverse Test");
  digitalWrite(left_reverse, HIGH);
  delay(delay_time_on);
  digitalWrite(left_reverse, LOW);
  delay(delay_time_off);
}
```


Feedback control

- we have 5 IR Sensors
 - Active means the IR is being reflected (happens with White)
 - lets encode each IR sensor as 0 or 1

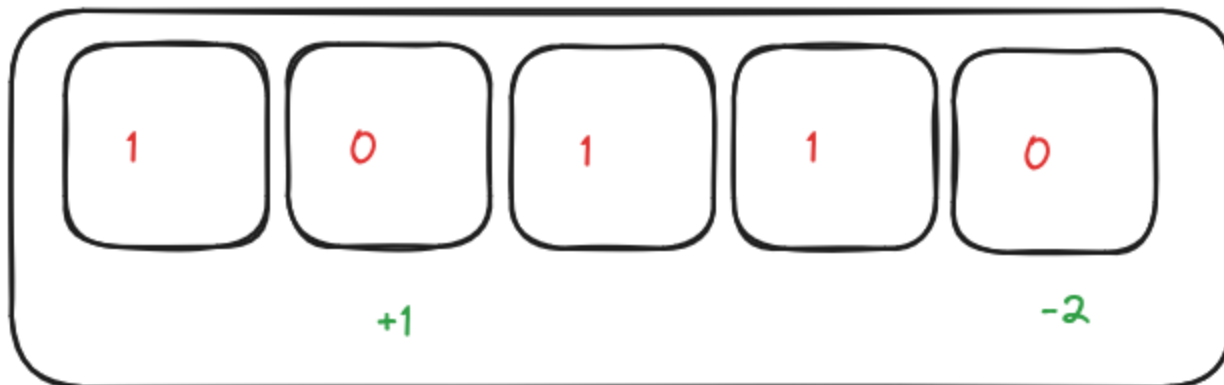
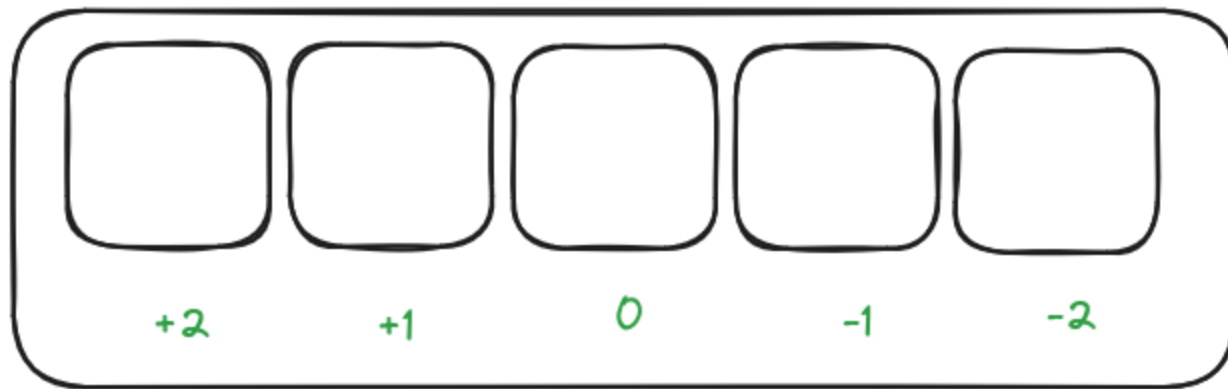




```
int lm = 5;  
int l1 = 18;  
int c = 19;  
int r1 = 22;  
int rm = 21;  
  
int irPins[] = {lm, l1, c, r1, rm};  
  
int pinRead[] = {0, 0, 0, 0, 0};
```

Defining Penalty System and Error Value

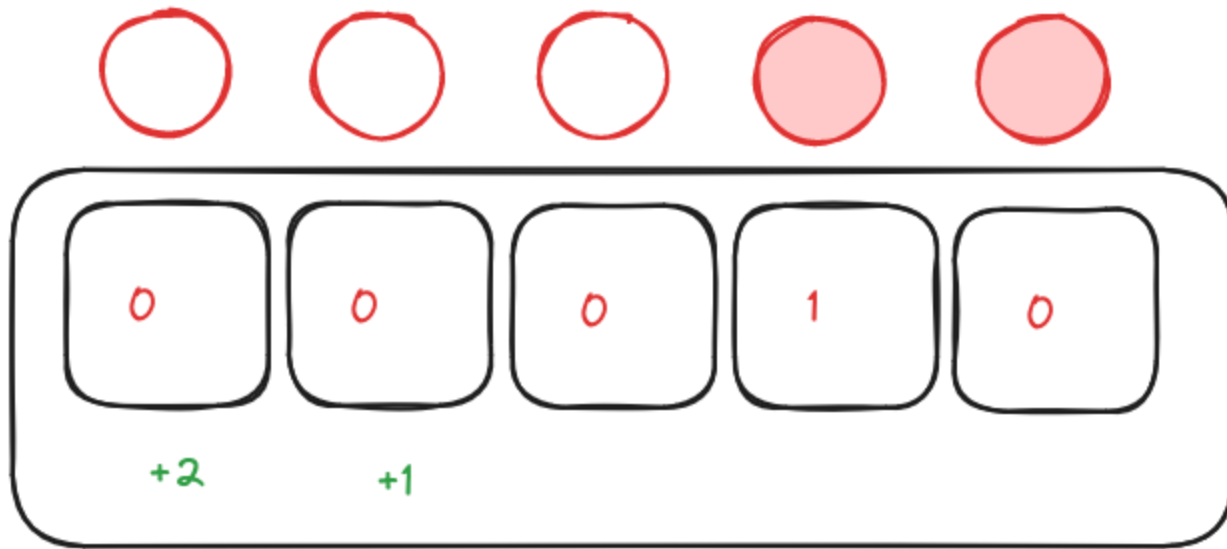
- First, when should we stop?
 - when everything is white 11111 (reflected)
 - when everything is black 00000 ?
 - there are definitely edge cases, which we will talk about
- when should we move forward, left , Right?
 - Lets design a penalty system based on Error

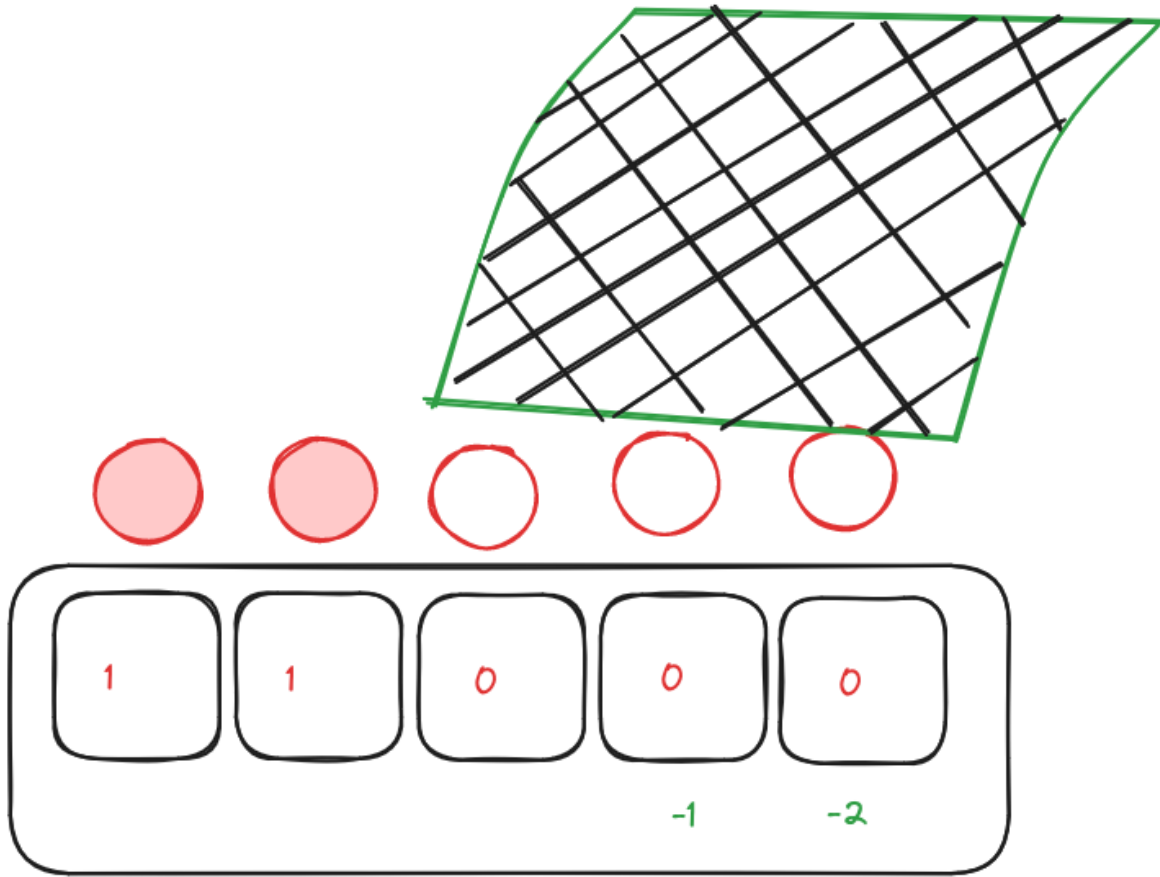


$$\text{error} = -2 + 1 = -1$$

Penalty

- if error is positive $error > 0$ then it means right wheel should move

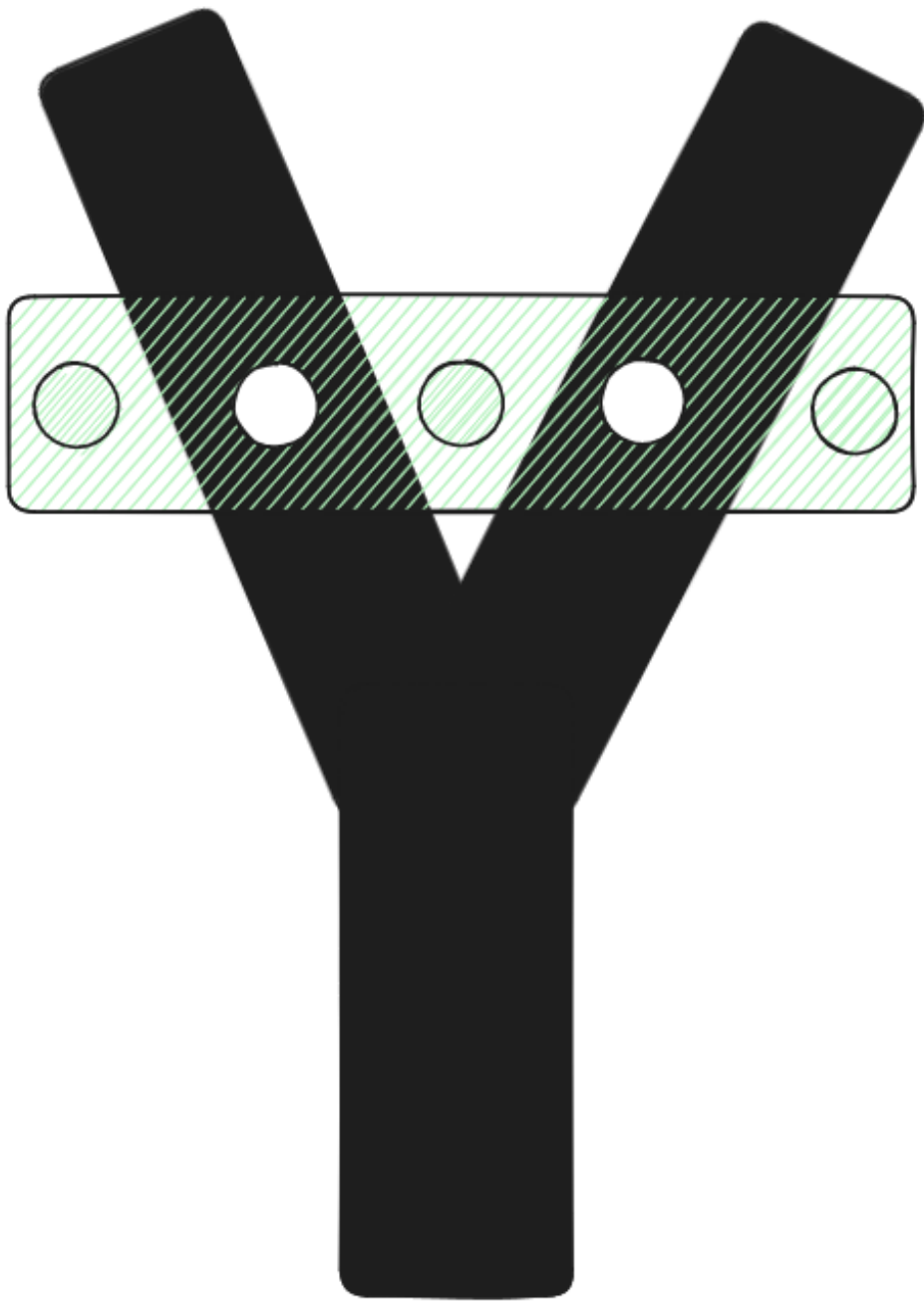




Left wheel should move, so car moves right
and follows the trail

Penalty

- if error is negative $error < 0$
then it means left wheel
should move, so the car moves
right

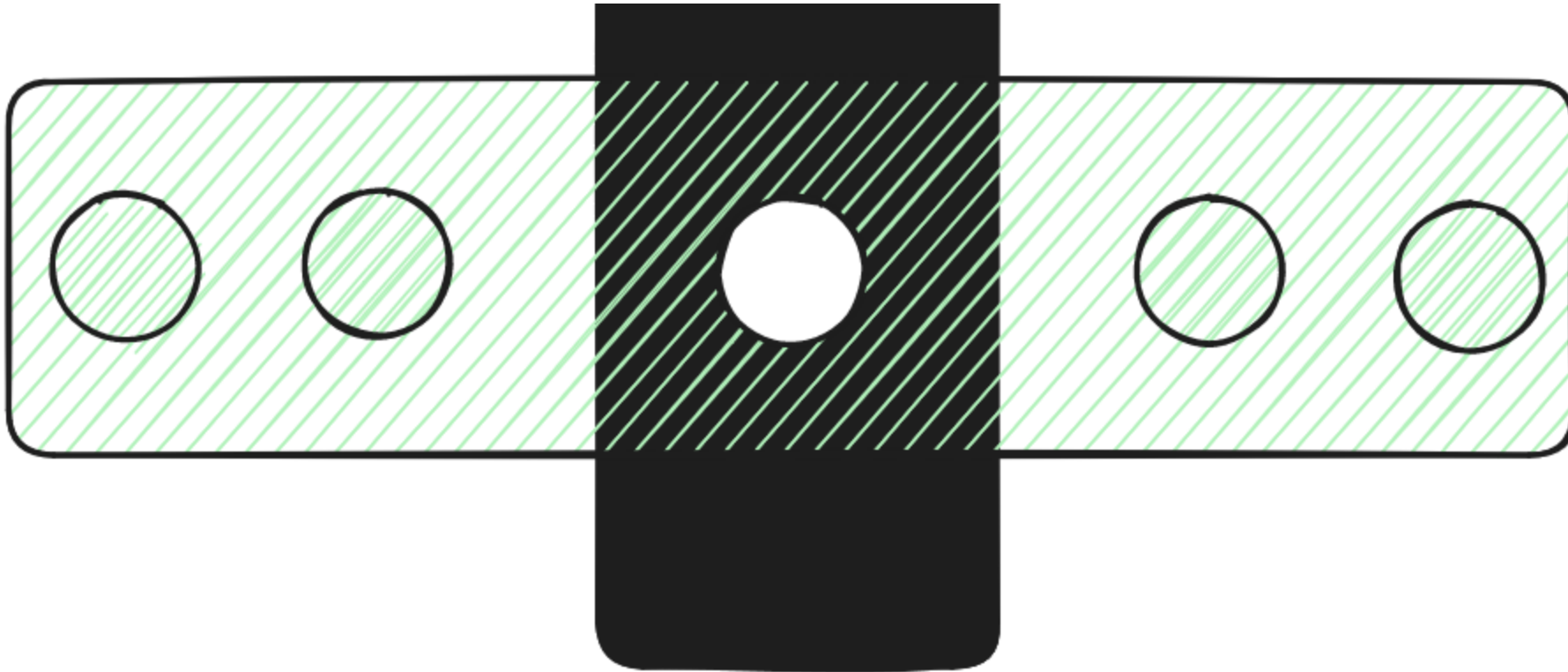


Penalty

- If Error is equal to zero
 - if sum of IR Sensor error values is equal to zero then you can either stop or choose left or right

Penalty

- If error is equal to zero, and other IR on Left and Right was not triggered then move forward



Final code

```
int right_forward = 28;
int right_reverse = 27;
int left_forward = 33;
int left_reverse = 32;
int delay_time_on = 50; // how long should each wheel turn?
int delay_time_off = 50;

int la = 5;
int lb = 30;
int c = 10;
int ra = 22;
int rb = 21;

int iRPins[] = {la, lb, c, ra, rb};

int pinHead[] = {0,0,0,0,0};
void setup() {
  pinMode(right_forward, OUTPUT);
  pinMode(right_reverse, OUTPUT);
  pinMode(left_forward, OUTPUT);
  pinMode(left_reverse, OUTPUT);
  // turn all the motors off
  digitalWrite(right_forward, LOW);
  digitalWrite(right_reverse, LOW);
  digitalWrite(left_forward, LOW);
  digitalWrite(left_reverse, LOW);
  pinMode(la, INPUT);
  pinMode(lb, INPUT);
  pinMode(c, INPUT);
  pinMode(ra, INPUT);
  pinMode(rb, INPUT);
  // for debugging, the output will appear on the serial monitor
  // To open the serial monitor, click the magnifying glass icon in the upper right corner
  Serial.begin(115200);
}

void loop() {
  for(int i = 0 ; i < 5; i++){
    pinHead[i] = digitalRead(iRPins[i]);
  }
  for(int i = 0 ; i < 5; i++){
    Serial.print(pinHead[i]);
    Serial.println("");
  }
  // Our error will be zero if the center IM is inactive
  // lets design a penalty based system where
  // if positive right wheel will move
  // if negative left wheel will move
  int error = 0;

  // la
  bool moveFlag = false;
  if (pinHead[0] == 0 ) { error += 2; moveFlag = true;}
  if (pinHead[1] == 0 ) { error += 1; moveFlag = true;}
  if (pinHead[3] == 0 ) { error -= 2; moveFlag = true;}
  if (pinHead[4] == 0 ) { error -= 2; moveFlag = true;}

  if(error > 0 ) { // "right wheel moves";
    rightWheelMoves();
  } else if(error < 0 ) {
    Serial.println("left wheel moves");
    leftWheelMoves();
  } else if(error == 0 ) {
    if (moveFlag == false && pinHead[2] == 0 ) {
      Serial.println("move forward");
      moveForward();
    } else {
      if (moveFlag) {
        if (pinHead[0] == 0 && pinHead[1] == 0 && pinHead[2] == 0 && pinHead[3] == 0 && pinHead[4] == 0 ) {
          Serial.println("stop");
          Stop();
        } else {
          Serial.println("move forward");
          moveForward();
        }
      } else {
        Serial.println("stop");
        Stop();
      }
    }
  }

  delay(50);
}

void leftWheelMoves() {
  digitalWrite(left_forward, HIGH);
  delay(delay_time_on);
  digitalWrite(left_forward, LOW);
  delay(delay_time_off);
}

void rightWheelMoves() {
  digitalWrite(right_forward, HIGH);
  delay(delay_time_on);
  digitalWrite(right_forward, LOW);
  delay(delay_time_off);
}

void moveForward() {
  digitalWrite(left_forward, HIGH);
  digitalWrite(right_forward, HIGH);
  delay(delay_time_on);
  digitalWrite(left_forward, LOW);
  digitalWrite(right_forward, LOW);
  delay(delay_time_off);
}

void Stop() {
  digitalWrite(right_forward, LOW);
  digitalWrite(right_reverse, LOW);
  digitalWrite(left_forward, LOW);
  digitalWrite(left_reverse, LOW);
}
```

What is PID

- most common form of feedback control
- there are PI control, PD control, etc too

PID controllers use 3 basic behaviour types of modes:

P – proportional

I – integral

D – derivative.

- Great for prototyping
- cross platform
- Has a very low power requirements
- Is open source with large community
- Doesn't require a degree worth of studying to program it.
- Very Cheap

What is possible with Arduino

- Insert Some projects here

Arduino Flavors

				 Lilypad
Arduino Uno	Arduino Mega	Arduino Nano	Arduino Leonardo	Arduino lilypad

Getting the Hardware

- If you are in Nepal
<https://himalayansolution.com/product/arduino-starter-kit-1>
- Access to International Payment then
<https://www.aliexpress.com/item/1005001653349193.html>

Stores in Nepal

- Supreme Light Technology: <https://www.sltech.com.np/>
- Himalayan Solution <https://himalayansolution.com/>

Microcontroller and Microprocessor

- Microcontroller is a chip that houses CPU, Memory(RAM ROM), I/O controllers all in a single package however Microprocessor however only has CPU
- A computer basic operation is `input -> process -> output`
- Atmega328P has 32kb Of program Memory, 1kb of ROM and 2Kb of RAM

Input	Process	Output

Arduino Board

Input	Output

Arduino Board Continued

For Input

- Analog Pins : Used to read analog sensor data. i.e Potentiometer, Pressure Sensor, Accelerometer, LDR sensor
- Digital Pins : If sensor already has A/D converter or the output of the sensor is discrete then Digital pins are used. i.e Ultrasonic Sensor, Ball Switch Sensor. It can only detect 0V or (3.3V/5V).

For Output

- PWM Pins : Pulse Width Modulation can be thought as a square wave continuously being on and off, where on time is the pulse width and it defines duty cycle.
- Digital Pins: They either provide 0V as off or 5V as On.

Arduino Board continued

Power Pins

test	ok
	• IOREF: Ioreference • RESET: Provide 5V to reset the board • 3.3V: 3.3V output for sensors • 5V: 5V output for sensors • GND: Ground Pin • VIN: Provide 5V to power the Arduino Uno Board

Basics of Electronics

- Ohm's law $V = IR$



- Voltage : Electric Pressure
- Current : Flow of electrons
- Resistance : Measure of opposition of current flow

Resistor



Exercise

Breadboard

- For Electronics Prototyping

Outside	Inside

Breadboard Continued

Breadboard Continued

Column1	Column2

LED (Light Emitting Diode)

<https://www.toppr.com/bytes/principles-of-led/>

- Active Semiconductor Device

Column1	Column2	Column3

LED

Series and Parallel Resistance

- Series Resistance $R_{eq} = R1 + R2 + R3 + R4$
- Parallel Resistance $\frac{1}{R_{eq}} = \frac{1}{R1} + \frac{1}{R2} + \frac{1}{R3} + \frac{1}{R4}$

Lets Build A Circuit

Series Resistance

<https://wokwi.com/projects/376191541801099265>

- Was there decrement in LED Glow?

Breadboard Path

Common Mistakes

Lets Build A Circuit

Parallel Resitance

<https://wokwi.com/projects/376191892088461313>

- Was there increment in LED Glow?

Potentiometer : Variabe Resitor

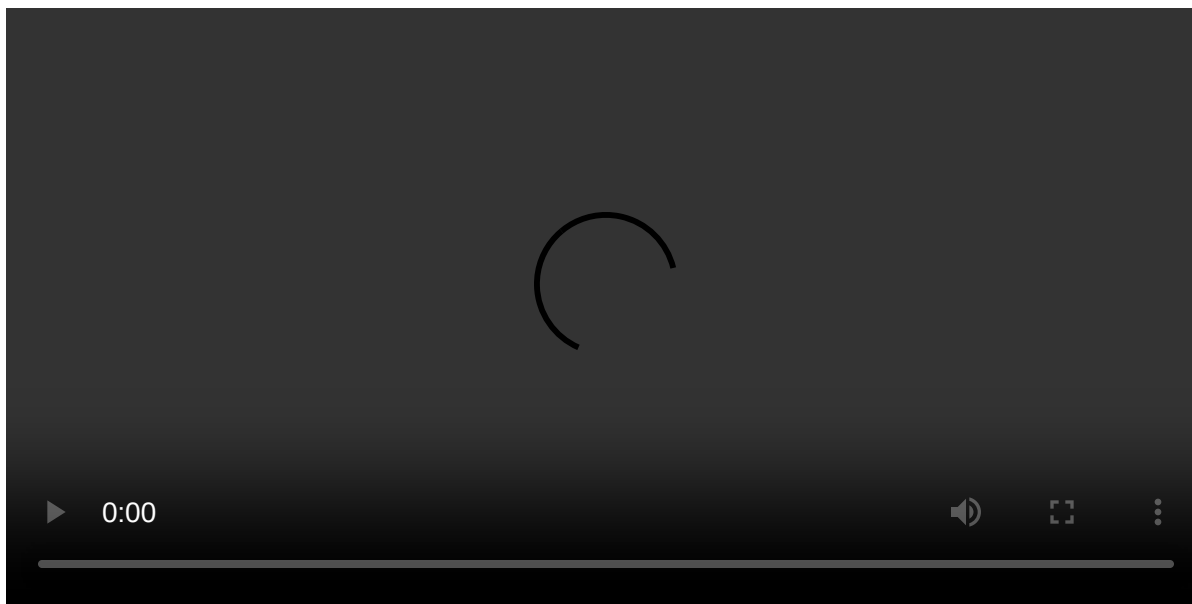
- $R = \rho \frac{L}{A}$

Build a circuit

- Potentiometer

<https://wokwi.com/projects/376209822976702465>

Potentiometer Video



Installing Arduino IDE

- For Linux : <https://docs.arduino.cc/software/ide-v1/tutorials/Linux>
- For Windows : <https://docs.arduino.cc/software/ide-v1/tutorials/Windows>
- For Mac : <https://www.arduino.cc/en/Guide/macOS>

Overview of writing code on Arduino IDE

- Write code in Arduino IDE
- Connect Arduino to PC using USB cable
- Select the Port
- Compile the Program
- click button on IDE to load the program
- Done

Write Some Code on Arduino IDE

General Structure

```
void setup() {  
    // setup code that runs once  
}  
  
void loop() {  
    // main code that runs repeatedly  
}
```

Example Code

Link to Simulator: <https://wokwi.com/projects/338430633664578130>

```
void setup() {  
    pinMode(5,OUTPUT); //set pin no 5 on OUTPUT mode  
}  
  
void loop() {  
    digitalWrite(5, HIGH); // send 5V  
    delay(1000); // wait for 1 sec  
    digitalWrite(5,LOW); // send 0 volts  
    delay(1000); // wait for 1 sec  
}
```

Example Code B

Link to Simulator: <https://wokwi.com/projects/338430886227739218>

```
void setup() {  
  pinMode(3, OUTPUT); // connected to PWM output  
}  
  
void loop() {  
  analogWrite(3,255); // to show how bright LED is at first  
  delay(1000); // later fades away  
  
  for(int i=0;i<255;i++){  
    analogWrite(3,i); // analogWrite(pin, value) ; where value can range from 0 to 255.  
    delay(100); // 0.1 sec delay  
  }  
}
```


Firmata

- Protocol for communicating with microcontrollers from software on a computer.
- All you need is to implement the protocol then any microcontroller can be supported.

To Upload Firmata

- connect the arduino to pc using USB cable
- open the arduino ide, select file>examples>firmata>StandardFirmata
- select port
- click upload

Firmata

Led Blink

```
var five = require("johnny-five");  
var board = new five.Board();  
  
board.on("ready", function() {  
    var led = new five.Led(13);  
    led.blink(500);  
});
```

PushButton

```
var five = require("johnny-five");
var board = new five.Board();

board.on("ready", function() {
  const led = new five.Led(7);
  const pushButton = new five.Button(
    {
      pin: 8,
      isPullup : true
    }
  );
  pushButton.on('down', () => {
    if (!led.isOn) {led.on();} else {led.off()}
  });
});
```

Analog Sensors

Analog Vs Digital

Analog Vs Digital

Aspect	Analog	Digital
Data Representation	Continuous waveform	Discrete binary values (0s and 1s)
Accuracy	High accuracy and precision	Excellent accuracy within bit resolution
Noise Resistance	Susceptible to noise and interference	More robust against noise
Signal Processing	Suitable for amplification, filtering, etc.	Versatile for complex algorithms
Cost and Complexity	Often simpler and potentially cheaper	Can be more complex and costly

LDR (Light Dependent Resistor)

<https://johnny-five.io/api/sensor/>

```
var five = require("johnny-five");

board = new five.Board();

board.on("ready", function() {

    let photoresistor = new five.Sensor({
        pin: "A0",
        freq: 500
    });

    let led = new five.Led(8);

    // board.repl.inject({
    //     pot: photoresistor
    // });

    photoresistor.on("data", function() {
        console.log(this.value);
        if (this.value >= 1000) {
            led.on();
        } else {
            led.off();
        }
    });
});
```

Equivalent C code

<https://wokwi.com/projects/376361780745305089>

```
#define LDR A0
#define LED 8

void setup() {
  pinMode(8, OUTPUT);
  pinMode(A0, INPUT);
}

void loop() {
  if (analogRead(A0) >= 1000) {
    digitalWrite(LED, HIGH);
  } else {
    digitalWrite(LED, LOW);
  }
}
```

Servo Control

Motors

Motor Type	Description	Image
Stepper	Converts electrical pulses into precise mechanical steps. Used in 3D printers and CNC machines.	
Servo	Provides accurate angular control, common in robotics.	
DC	Operates on direct current, simple, and used in various applications.	

Servo Motor

```
const five = require("johnny-five");
const board = new five.Board();

board.on("ready", () => {
  const servo = new five.Servo(10);

  board.repl.inject({
    servo
  });
  servo.sweep();
})
```

Build a project

LED Binary Counter

Circuit

<https://wokwi.com/projects/376380254565660673>


```

const five = require("johnny-five");
const board = new five.Board();

board.on("ready", () => {
  let leds = [8,9,10,11,12].map((pinNo) => {
    return new five.Led(pinNo);
  });

  for (let i = 0 ; i <= 31; i++) {
    setTimeout(() => {ledState(i.toString(2).padStart(5, '0') , leds)}, i * 1000);
  }

});

function ledState(ledString, leds) {
  Array.from("00000").map((val, index) => setLedState(leds[index], parseInt(val)));
  Array.from(ledString).map((val, index) => setLedState(leds[index], parseInt(val)));
}

function setLedState(led, state) {
  state == 1 ? led.on() : led.off();
}

```

Robomaze and Line followig robot

Micromouse

Fastest Maze-Solving Competition on Earth

Line following Robot

Components Required

- Chassis
- 9v battery
- Arduino Uno
- L293 Motor driver
- IR Sensor

Thankyou

